MIT 8103 — Advanced Database Systems

Community Library Management System Portfolio

Course: MIT 8103 Advanced Database Systems
Case study: Community Library Management System

This repository contains five portfolio activities, Python scripts for the SQLite implementation and testing, and supporting evidence from the local execution and benchmarking work. The relational database implementation and local SQLite checks are executable. MongoDB commands are illustrative examples only and have not been executed against a MongoDB server. The distributed/cloud architecture is a design proposal and has not been deployed.

Repository structure

```
README.md

portfolio-1-database-design/
    normalisation.md

portfolio-2-query-optimisation/
    analysis_notes.md

portfolio-3-transactions-concurrency/
    concurrency-notes.md
    issue_book.py

portfolio-4-nosql/
    loan_document.json
    mongodb_examples.js

portfolio-5-distributed-cloud/
    architecture.md

scripts/
    build_database.py
    issue_book.py
    run_all_checks.py
    run_query_analysis.py
    test_transactions.py

evidence/
    benchmark_dataset_profile.json
    benchmark_results.json
    benchmark_summary.csv
    build_validation.json
    build_validation.txt
    query_analysis_console.txt
    query_analysis_interpretation
    query_plans_before_after.txt
    reproduction_run.txt
    transaction_test_output.txt
```

Requirements

• Python 3.10 or later
• SQLite, available through Python's built-in sqlite3 module
• No third-party Python packages are required for the supplied scripts.

Reproducing the SQLite work

Run commands from the repository root directory:

```
python scripts/run_all_checks.py
```

This runs the database build and validation, transaction tests and synthetic query benchmark. Individual scripts can also be run:

```
python scripts/build_database.py
python scripts/test_transactions.py
python scripts/run_query_analysis.py
```

The benchmark uses a deterministic synthetic dataset consisting of 1,000 books, 1,500 copies, 3,000 members, 8,000 loan headers and 12,000 loan-item history rows. Four query patterns are warmed up and timed 15 times before and after indexing. Median timings and EXPLAIN QUERY PLAN output are recorded. Results depend on the execution environment and are not production performance guarantees.

Generated database files are created locally when the scripts are run; they are not included as committed database files in this repository.

Evidence and execution status

Component	Status
Relational database implementation and local validation	Executed locally
Four SQL query patterns, query plans and synthetic benchmark	Executed locally
Issue transaction commit, rejection and rollback tests	Executed locally
Two-session concurrent writer scenario	Documented test plan; not executed
MongoDB loan document and mongosh examples	Examples prepared; not executed against MongoDB
Distributed/cloud architecture	Design proposal only; no cloud deployment

The evidence/ directory contains the benchmark outputs, database validation records, query-plan comparisons, reproduction log and transaction-test output. All people and email addresses in the sample data are synthetic.

AI-use declaration

AI assistance was used to help organise, draft and review parts of this portfolio. The student is responsible for reviewing and understanding the submitted work, verifying its contents and ensuring that this declaration accurately reflects the assistance received.